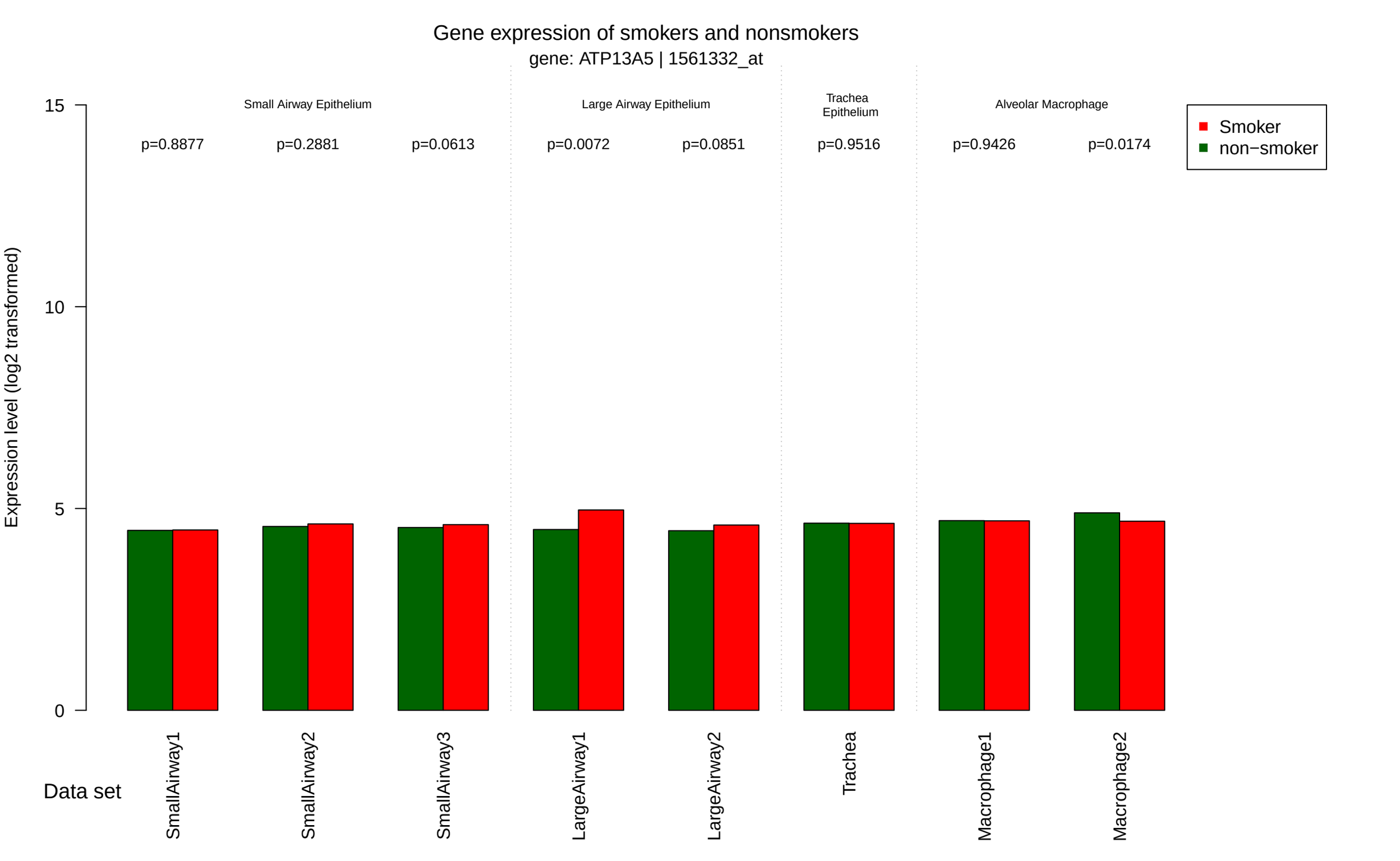
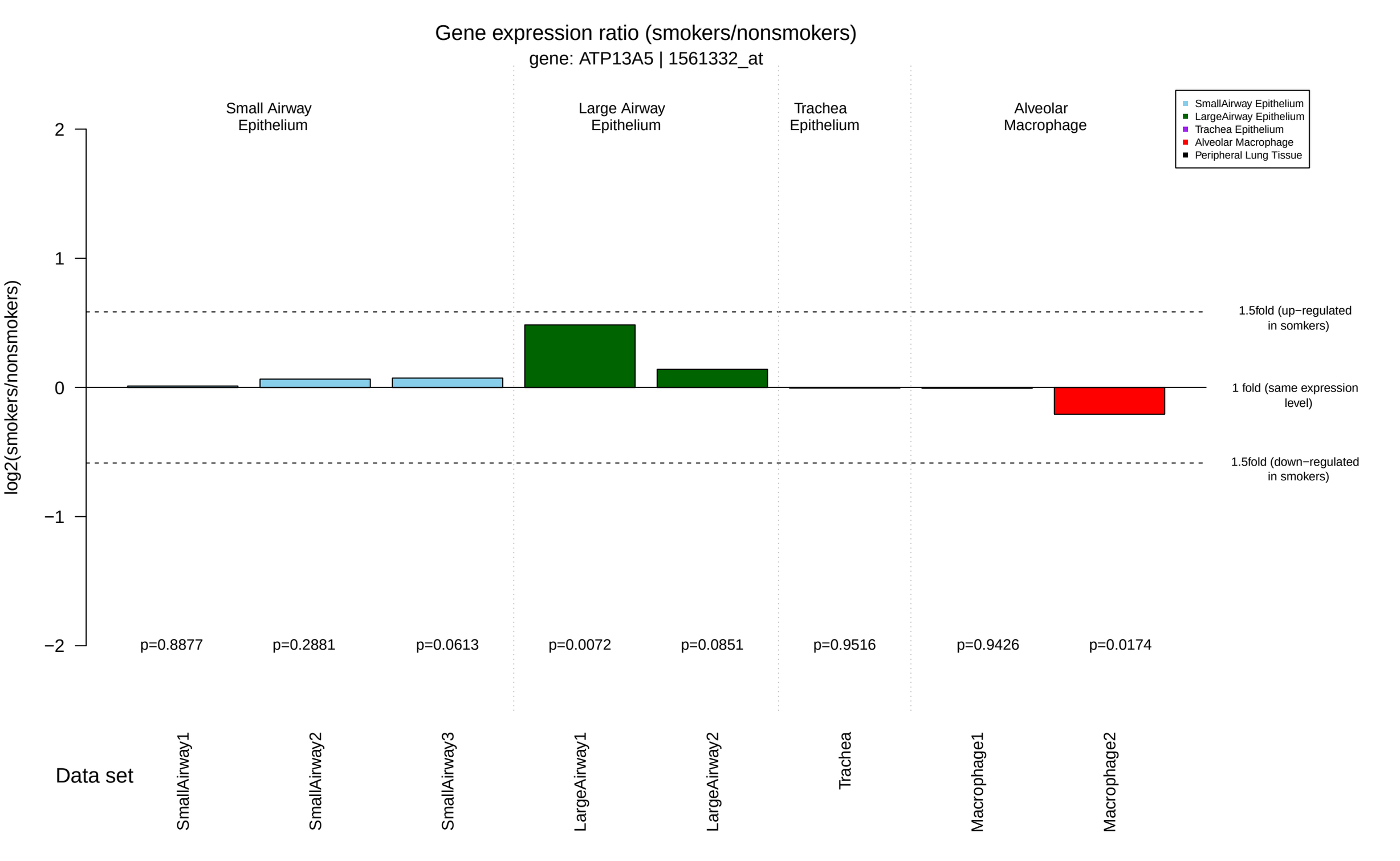
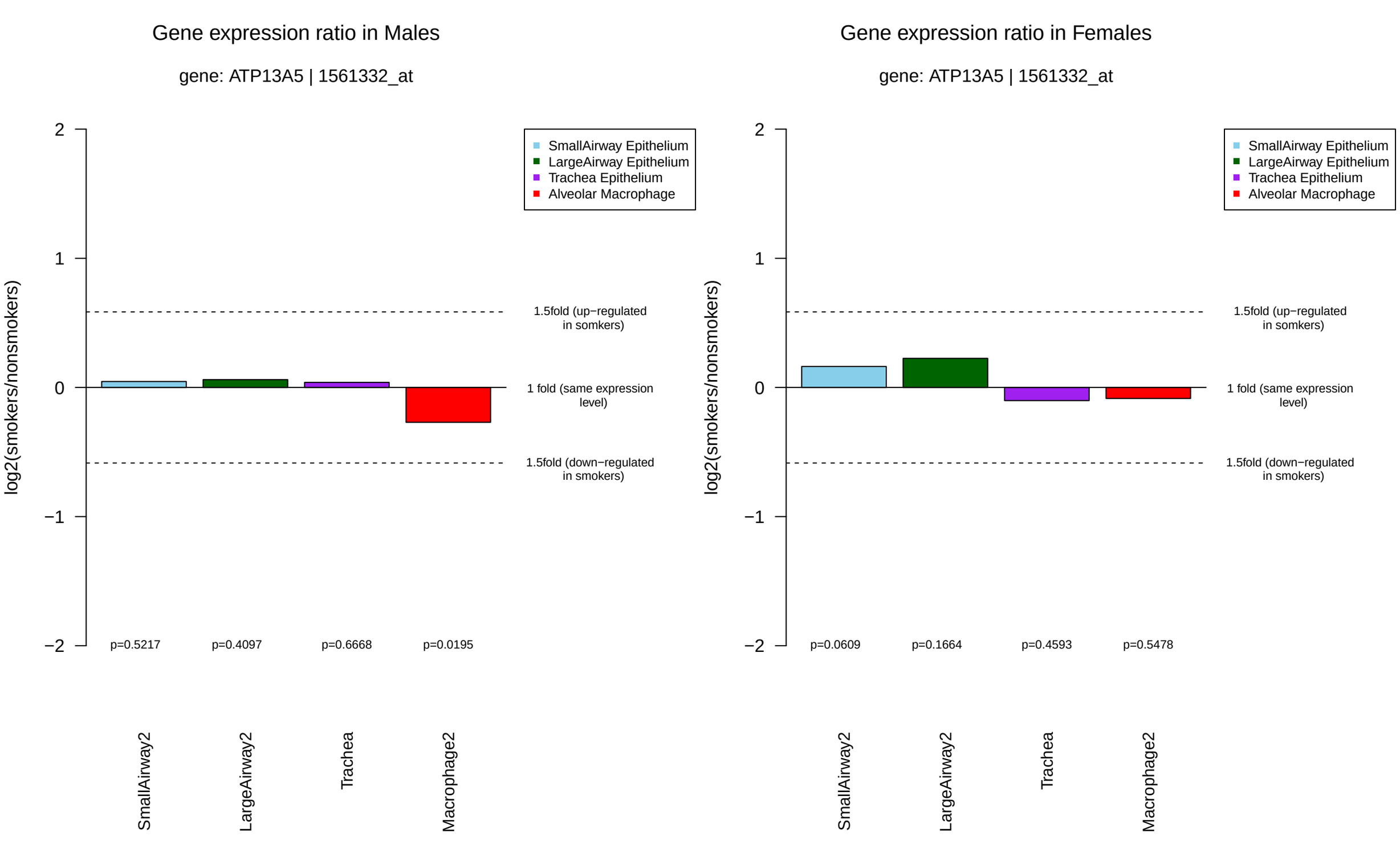
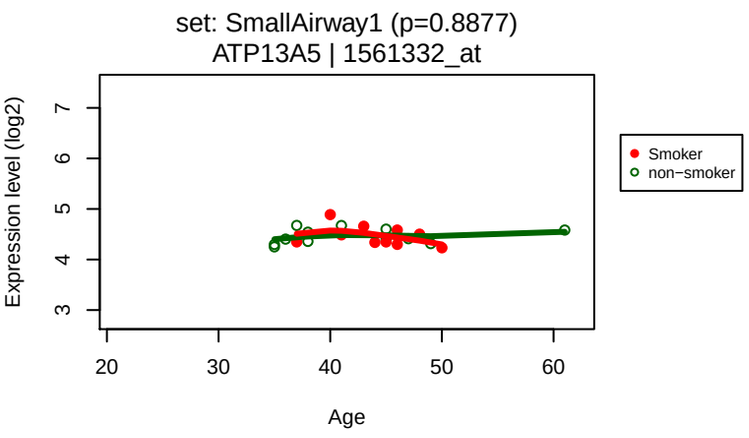
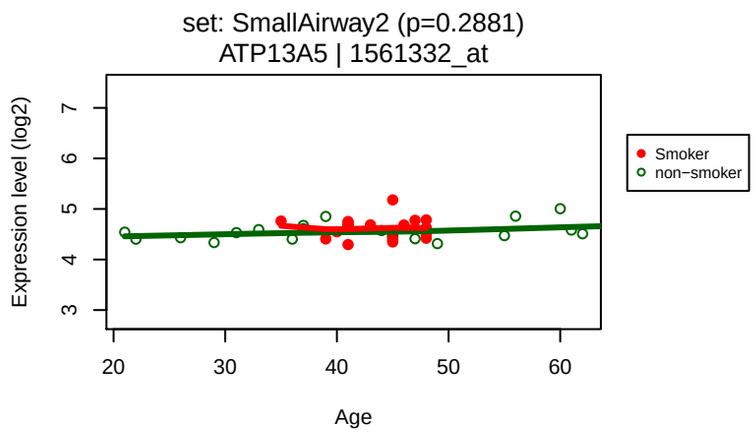
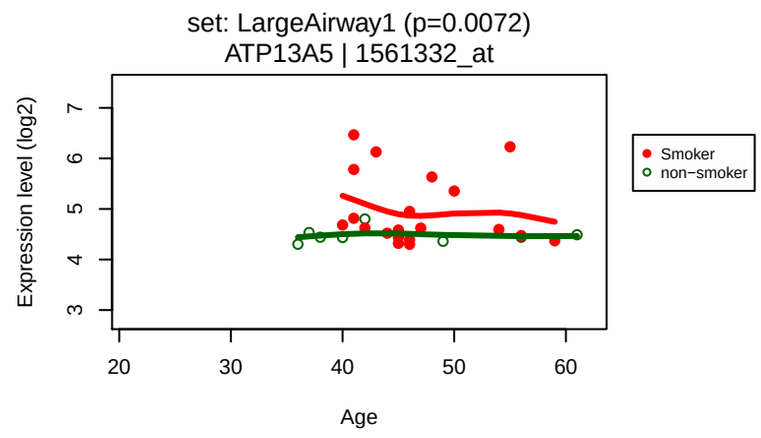

Gene expression of smokers and nonsmokers
gene: ATP13A5 | 1561332_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.8877
p=0.2881
p=0.0613
p=0.0072
p=0.0851
p=0.9516
p=0.9426
p=0.0174
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

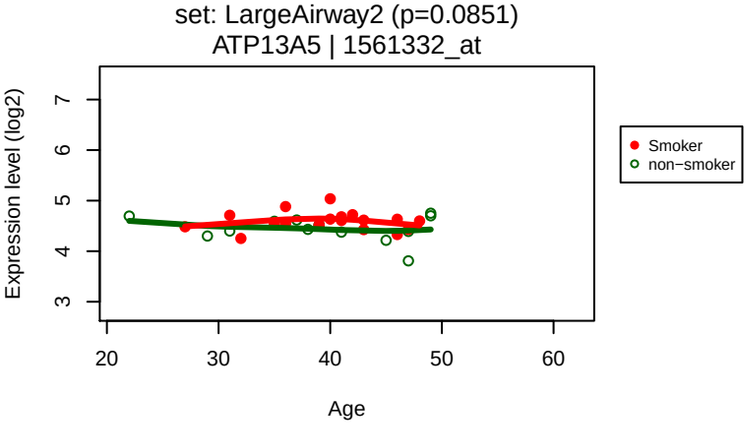

Gene expression ratio (smokers/nonsmokers)
gene: ATP13A5 | 1561332_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8877
p=0.2881
p=0.0613
p=0.0072
p=0.0851
p=0.9516
p=0.9426
p=0.0174
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

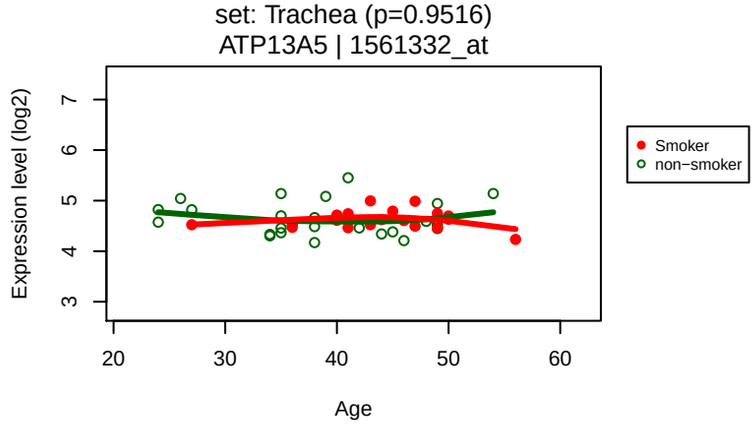

Gene expression ratio in Males
gene: ATP13A5 | 1561332_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5217
p=0.4097
p=0.6668
p=0.0195
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ATP13A5 | 1561332_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0609
p=0.1664
p=0.4593
p=0.5478
SmallAirway2
LargeAirway2
Trachea
Macrophage2

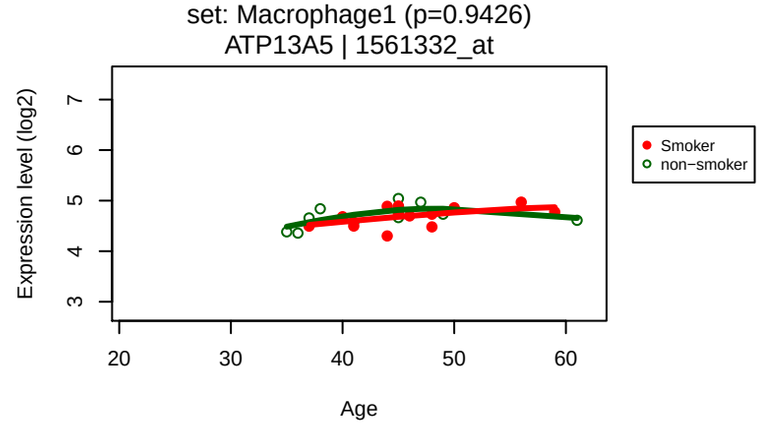

set: SmallAirway1 (p=0.8877)
ATP13A5 | 1561332_at
Expression level (log2)
Age
Smoker
non−smoker

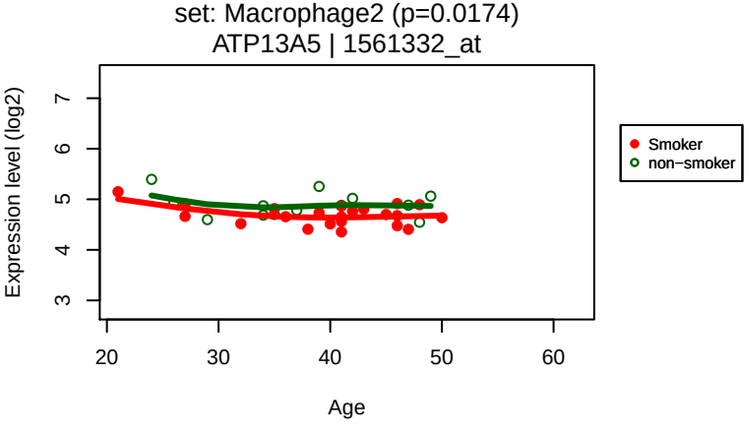

set: SmallAirway2 (p=0.2881)
ATP13A5 | 1561332_at
Expression level (log2)
Age
Smoker
non−smoker

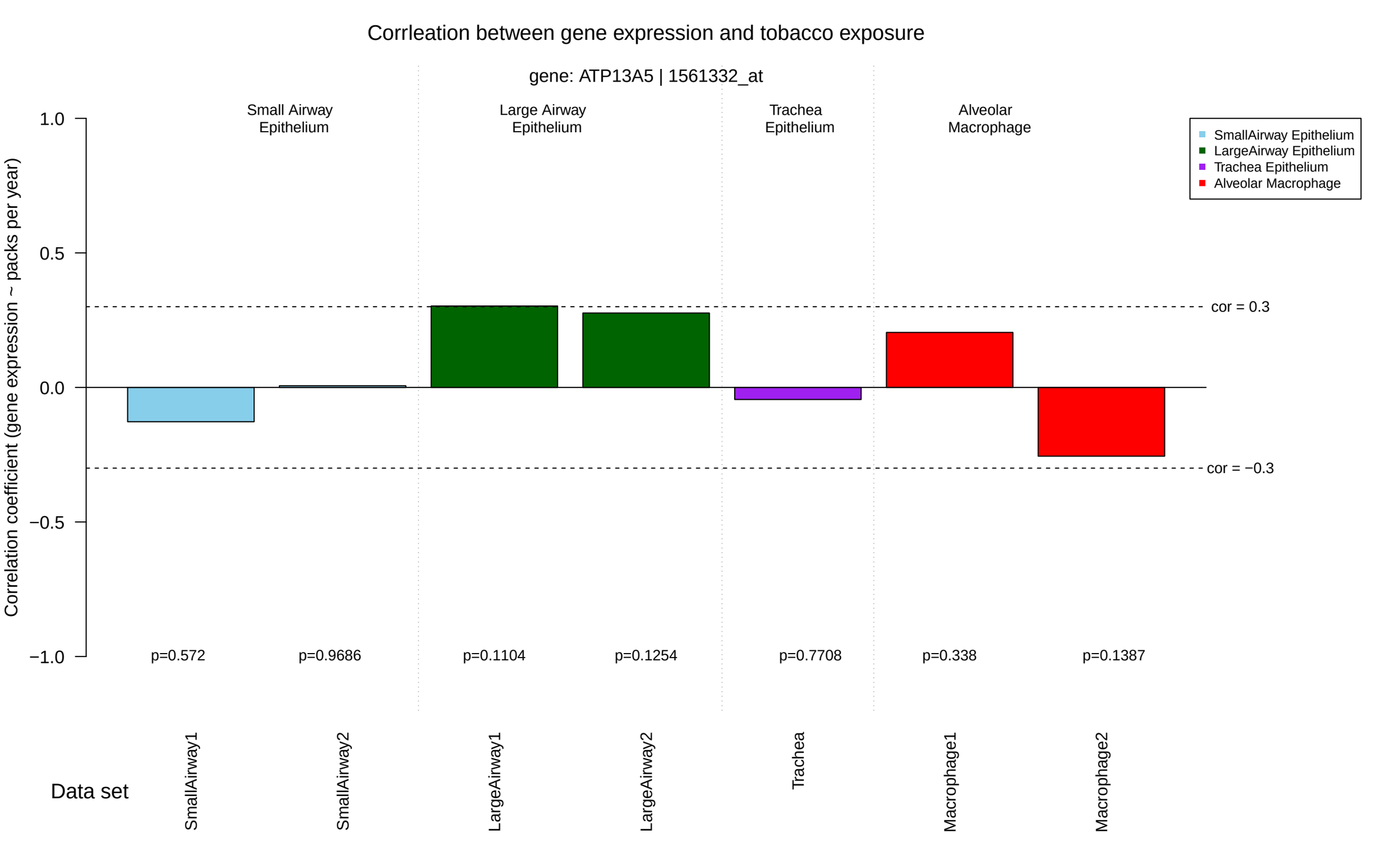

set: LargeAirway1 (p=0.0072)
ATP13A5 | 1561332_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0851)
ATP13A5 | 1561332_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.9516)
ATP13A5 | 1561332_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9426)
ATP13A5 | 1561332_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0174)
ATP13A5 | 1561332_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ATP13A5 | 1561332_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.572
p=0.9686
p=0.1104
p=0.1254
p=0.7708
p=0.338
p=0.1387
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2